

# CS & IT ENGINEERING

Programming in C

**Arrays and Pointers**

**Lec- 04**



By- Pankaj Sharma sir





TOPICS TO BE  
COVERED



Pointers

```
int a[4] = {10, 20, 30, 40};
```

```
int *p;
```

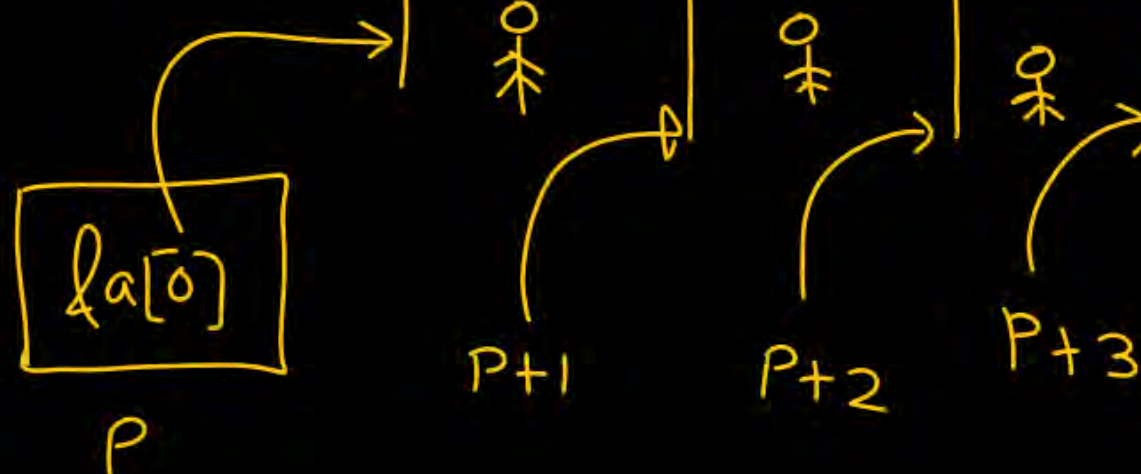
```
p = &a[0];
```

```
printf("/d", *(p+0)); 10  
printf("/d", *(p+1)); 20  
printf("/d", *(p+2)); 30  
printf("/d", *(p+3)); 40
```

```
for(i=0; i<4; i++)
```

```
printf("/d", *(p+i));
```

| a[0] | a[1] | a[2] | a[3] |
|------|------|------|------|
| 10   | 20   | 30   | 40   |



$$P+0 = \&a[0]$$

$$*(P+0) = \cancel{\&a[0]} = a[0]$$

$$P = \&a[0]$$

$$P+2 = \&a[0] + 2 = \&a[2]$$

$$*(P+2) = \cancel{\&a[2]} = a[2]$$

$$P = \&a[0]$$

$$P+1 = \&a[0] + 1 = \&a[1]$$

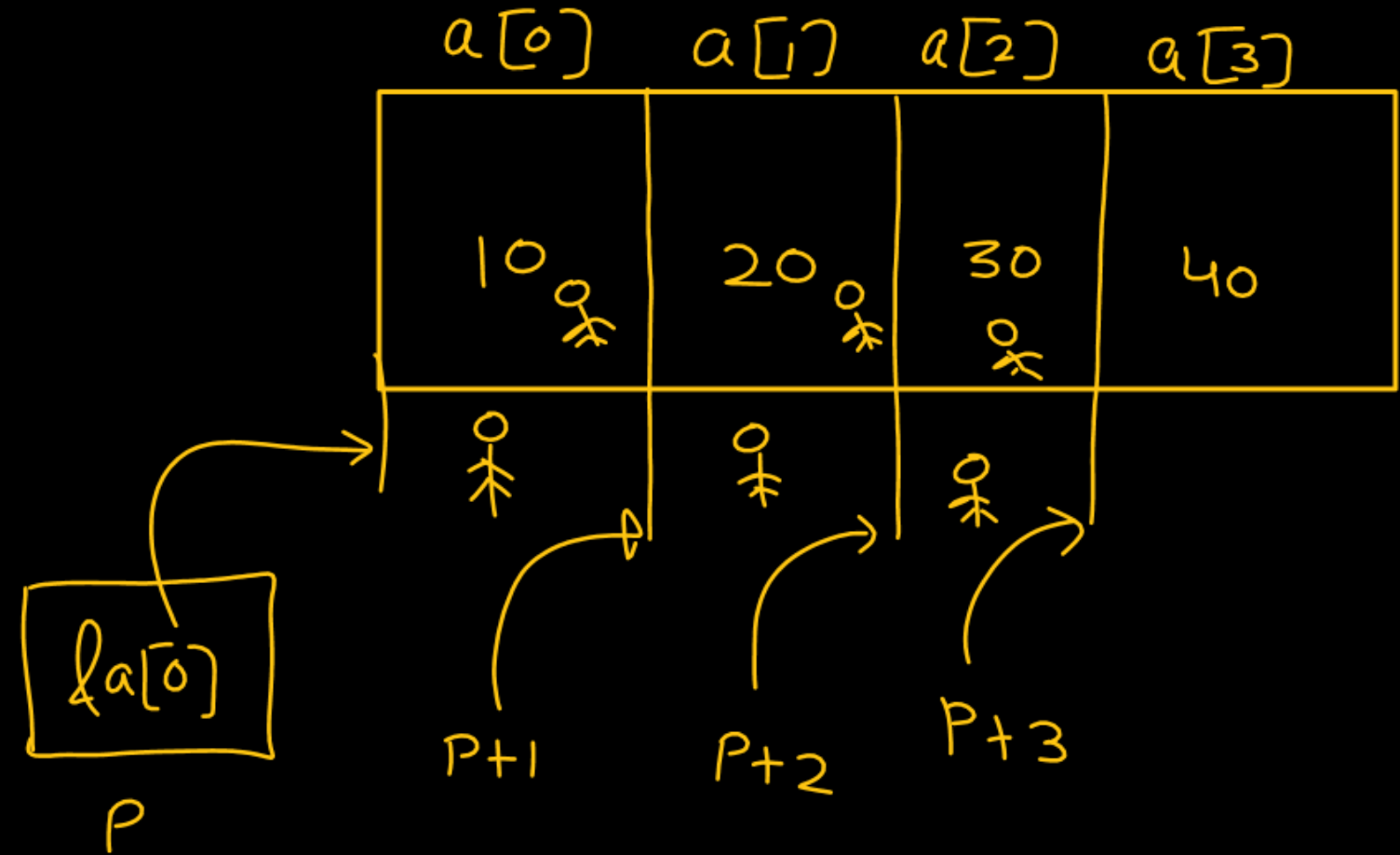
$$*(P+1) = \cancel{\&a[1]} = a[1]$$

```
int a[4] = {10, 20, 30, 40};
```

```
int *p;
```

```
p = &a[0];
```

```
for (i = 0; i < 4; i++)  
    printf("%d", p[i]);
```



```
int a[4] = {10, 20, 30, 40};
```

```
a++;  
++a;  
--a;  
a--;
```

Invalid

Lvalue

```
a = { };
```

Invalid

```
int a[4] = {10, 20, 30, 40};
```

```
int* p;
```

```
p = &a[1];
```

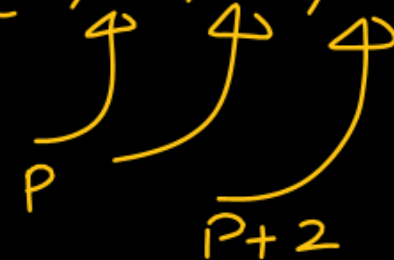
```
++p; ✓
```

```
--p; ✓
```

```
p++; ✓
```

```
p--; ✓
```

```
p = p + 2; ✓
```





```
# void main() {  
    int a[4] = { 10, 20, 30, 40 };
```

Valid Program

```
        12;  
        13.48;  
    }
```

# \_\_\_\_\_  
void main() {

int a[4] = {10, 20, 30, 40};

int \*p;

p = &a[0];

p++;

printf("%d", \*p);

}

Will not  
update the  
value of  
p

# \_\_\_\_\_  
void main() {

int a[4] = {10, 20, 30, 40};

int \*p;

p = &a[0];

p = p++;

printf("%d", \*p);

}

updating p

```
#include <stdio.h>
```

```
void main(){
```

```
int a[4] = {10, 20, 30, 40};
```

```
int *p;
```

```
p = &a[0];
```

```
printf("/d", *++p);
```

```
++*p;
```

```
++p;
```

```
printf("/d", *p++);
```

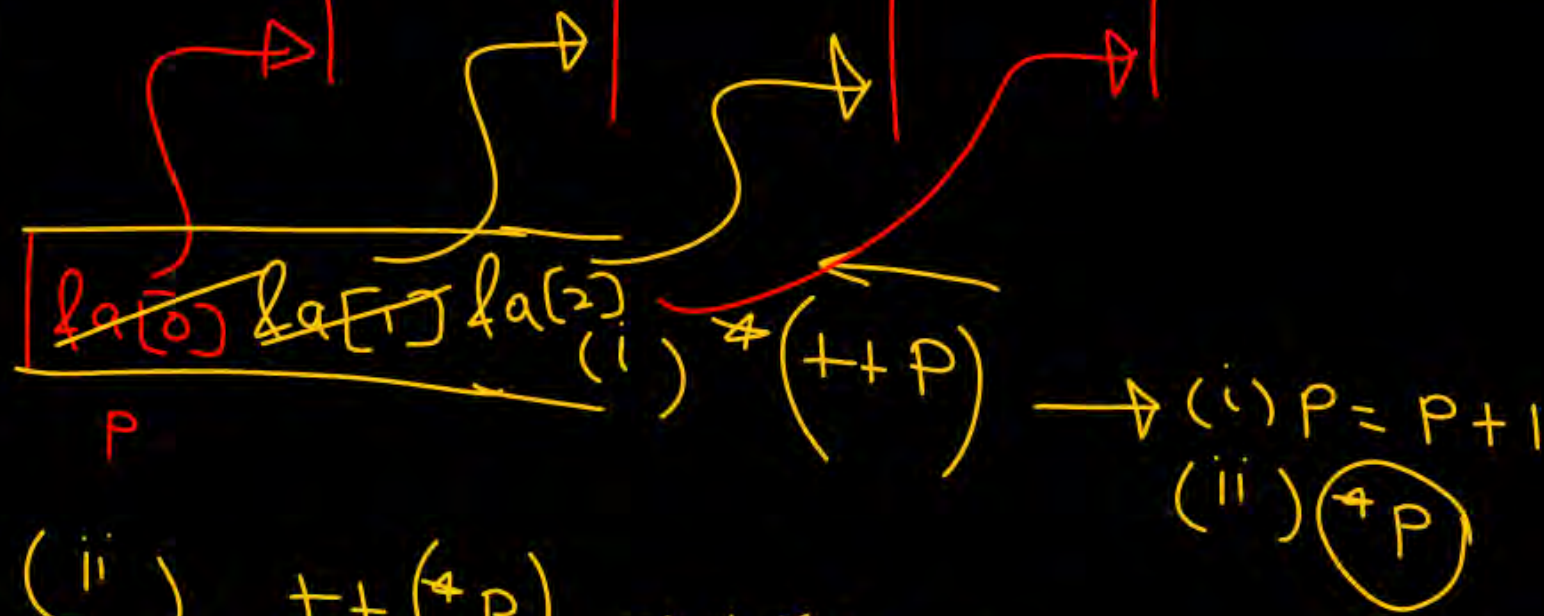
```
}
```

(i) \*p (pf)  
(ii) p = p + 1

20, 30

++, \* Same  
R to L

| a[0] | a[1]          | a[2] | a[3] |
|------|---------------|------|------|
| 10   | <del>20</del> | 30   | 40   |



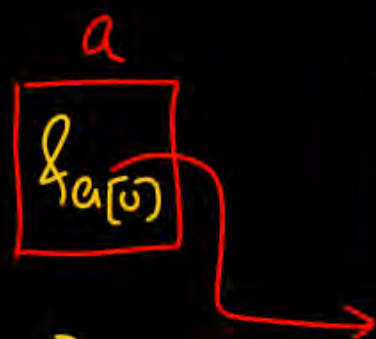
(ii) ++(\*p) ⇒ (i) \*p = \*p + 1  
(ii) use \*p X



$a = \&a[0]$   
 $a+2 \Rightarrow \&a[2]$

`int a[4] = {10, 20, 30, 40};`

`int *p[4] = {a+2, a, a+1, a+3};`



| $a[0]$ | $a[1]$ | $a[2]$ | $a[3]$ |
|--------|--------|--------|--------|
| 10     | 20     | 30     | 40     |

`int **q;`

`q = \&p[0];`

`printf("/d", *++*++q);`



|          |          |          |          |
|----------|----------|----------|----------|
| $\&a[2]$ | $\&a[1]$ | $\&a[1]$ | $\&a[3]$ |
| $p[0]$   | $p[1]$   | $p[2]$   | $p[3]$   |

`*++(*++q)`



(i)  $q = q+1 \Rightarrow q = \&p[0]+1 = \&p[1]$

$*++(*++q)$

(i)  $*q = *q+1$   
 $p[1] = p[0]+1$

(ii)  $**q = \&a[0]+1 = \&a[1]$

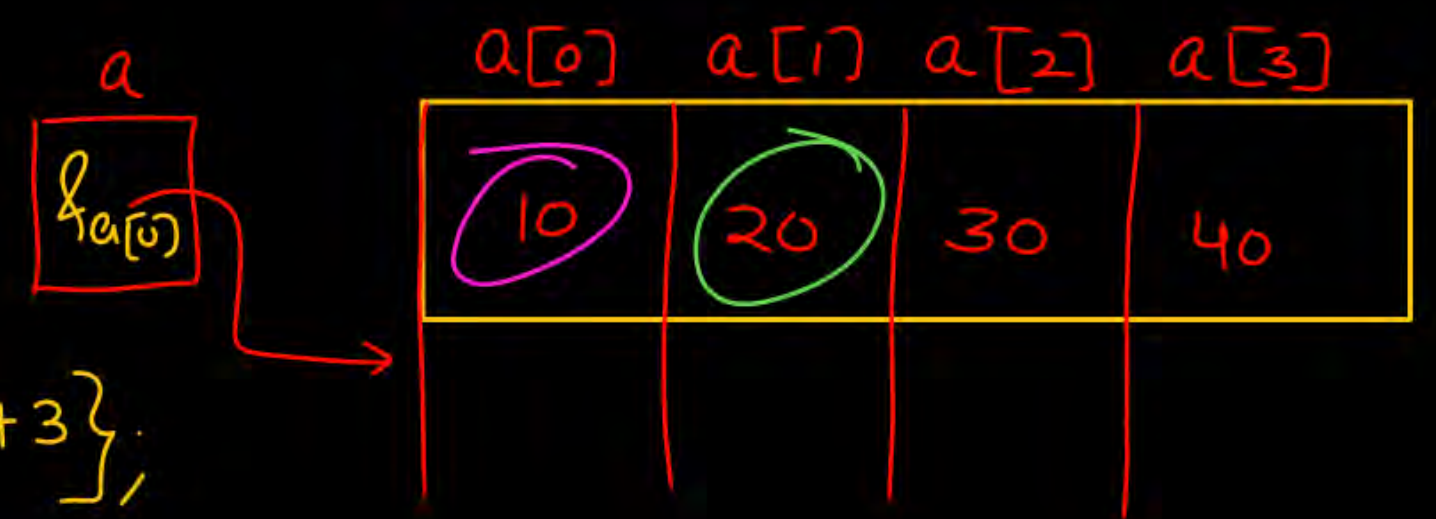
$\Rightarrow *q = \&p[1]$   
 $\Rightarrow *p[1] \Rightarrow \&a[1]$



$a = \&a[0]$   
 $a+2 \Rightarrow \&a[2]$

`int a[4] = {10, 20, 30, 40};`

`int *p[4] = {a+2, a, a+1, a+3};`



`int **q;`

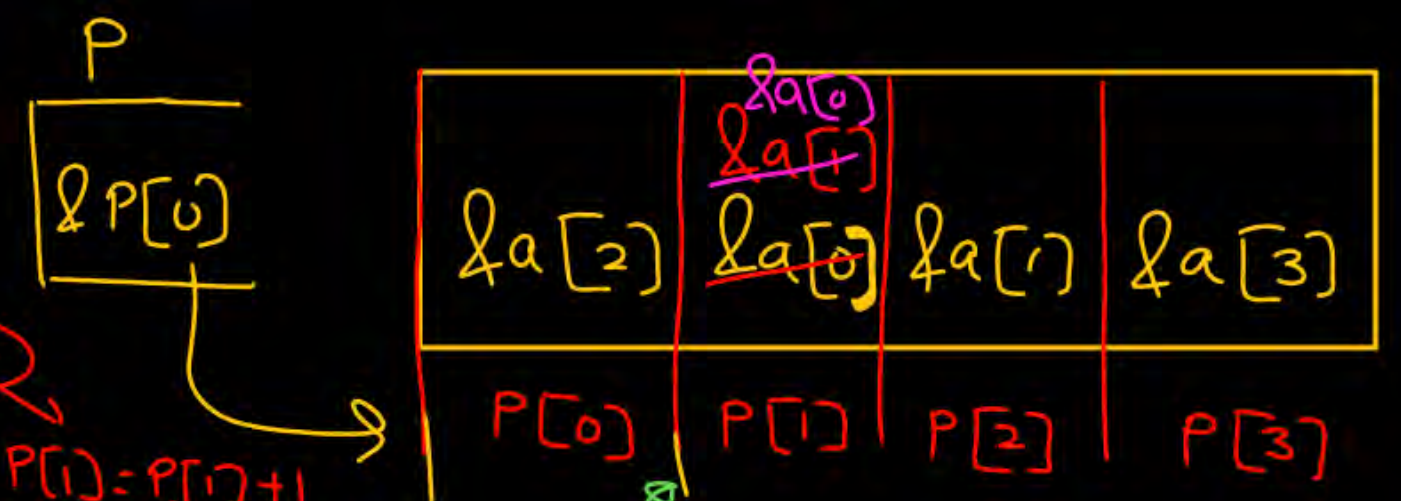
`q = \&p[0];`

`*++(*++q);`  
`--q;`

`printf("/d", *--*++q);`

(i)  $++q$

(ii)  $*--*q \Rightarrow q$   
 $a) \&q = \&q-1$   
 $b) **q$



~~$\&p[0]$~~   ~~$\&p[1]$~~   ~~$\&p[2]$~~   $\&p[3]$   
 $p[1] = p[1] + 1 = \&a[0] + 1$

$p[1] = p[1] - 1 = \&a[1] - 1 = \&a[0]$

~~$**\&p[1]$~~   
 ~~$*p[1]$~~   
 ~~$*\&a[0]$~~   
 $\Rightarrow a[0]$

```
# void fun(int*);
void main(){
```

```
int a[4] = {10, 20, 30, 40};
```

```
fun(a);
```

```
printf("%d %d", a[0], a[1]);
}
```

```
void fun(int* p){
```

```
++p;
```

```
*p++;
```

```
++*p;
}
```

a[0] a[1] a[2] a[3]

|    |    |                     |    |
|----|----|---------------------|----|
| 10 | 20 | <del>30</del><br>31 | 40 |
|----|----|---------------------|----|



\*p++

\*(p++)

(i) (\*p) ✓

(ii) p = p + 1; ✓

\*(p++) → (i) (\*p) ×  
(ii) p = p + 1

++(\*p) ⇒ (i) \*p = \*p + 1  
(ii) \*p ×



```
# void fun(int*);  
void main(){
```

```
    int a[4] = {10, 20, 30, 40};
```

```
    fun(a);
```

```
    printf("/d /d", a[0], a[1]);  
}
```

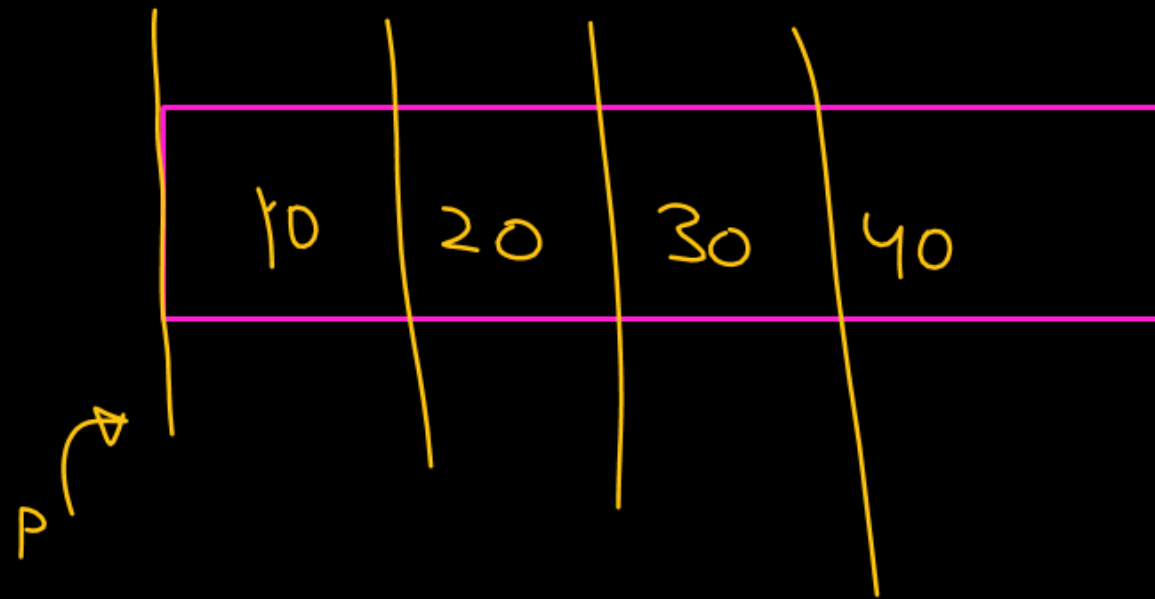
```
void fun(int *p){  
    ++p;  
    *p++;  
    ++*p;  
}
```

void fun(int P[])

int \*p

}

```
void main(){  
    int a[4] = {10, 20, 30, 40};  
  
    fun(a, 4);  
}
```



```
void fun(int *p, int n)  
{  
    int i, sum = 0;  
  
    for(i = 0; i < n; i++)  
    {  
        sum = sum + p[i];  
    }  
    printf("sum is %d", sum);  
}
```



```
void main(){
```

```
int a[3][3] = { {1,2,3}, {4,5,6}, {7,8,9} };
```

```
fun(a);
```

```
printf("/d /d /d", a[1][1], a[1][2], a[2][0]);
}
```

```
void fun(int (*P)[3])
{
```

```
(*P)[1] = (*P)[1] + 1;
```

```
}
```

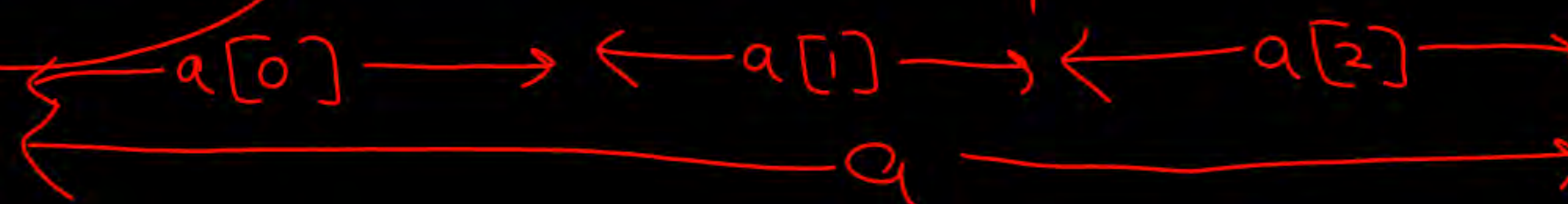
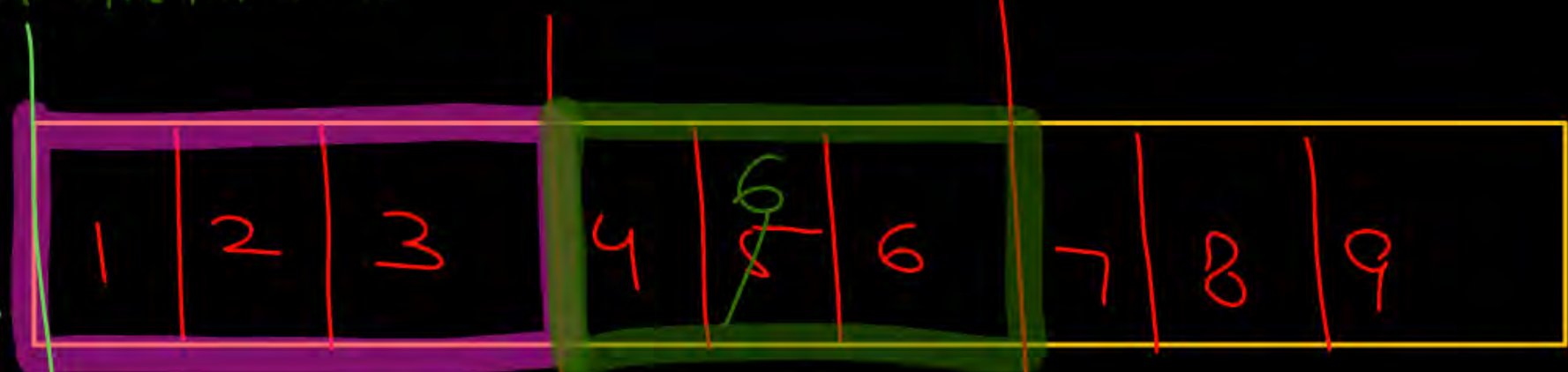
$(*P)[1]$

$(*a[1])[1]$

$\Rightarrow a[1][1]$

Read the declaration

$P = \&a[0]$   
 $++P \Rightarrow P = \&a[1]$

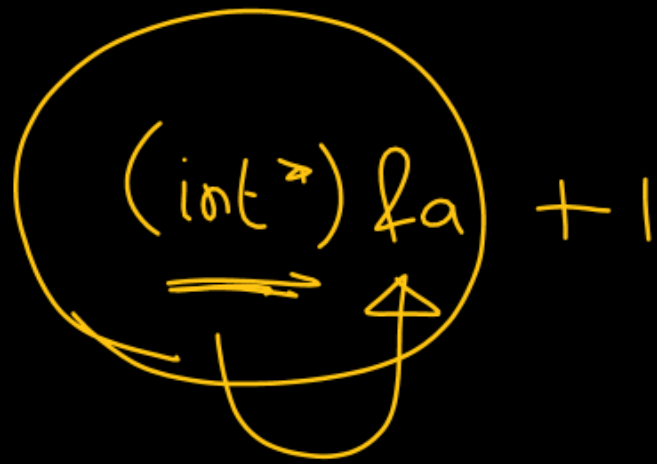


int a[4] = {10, 20, 30, 40};

int \*p;

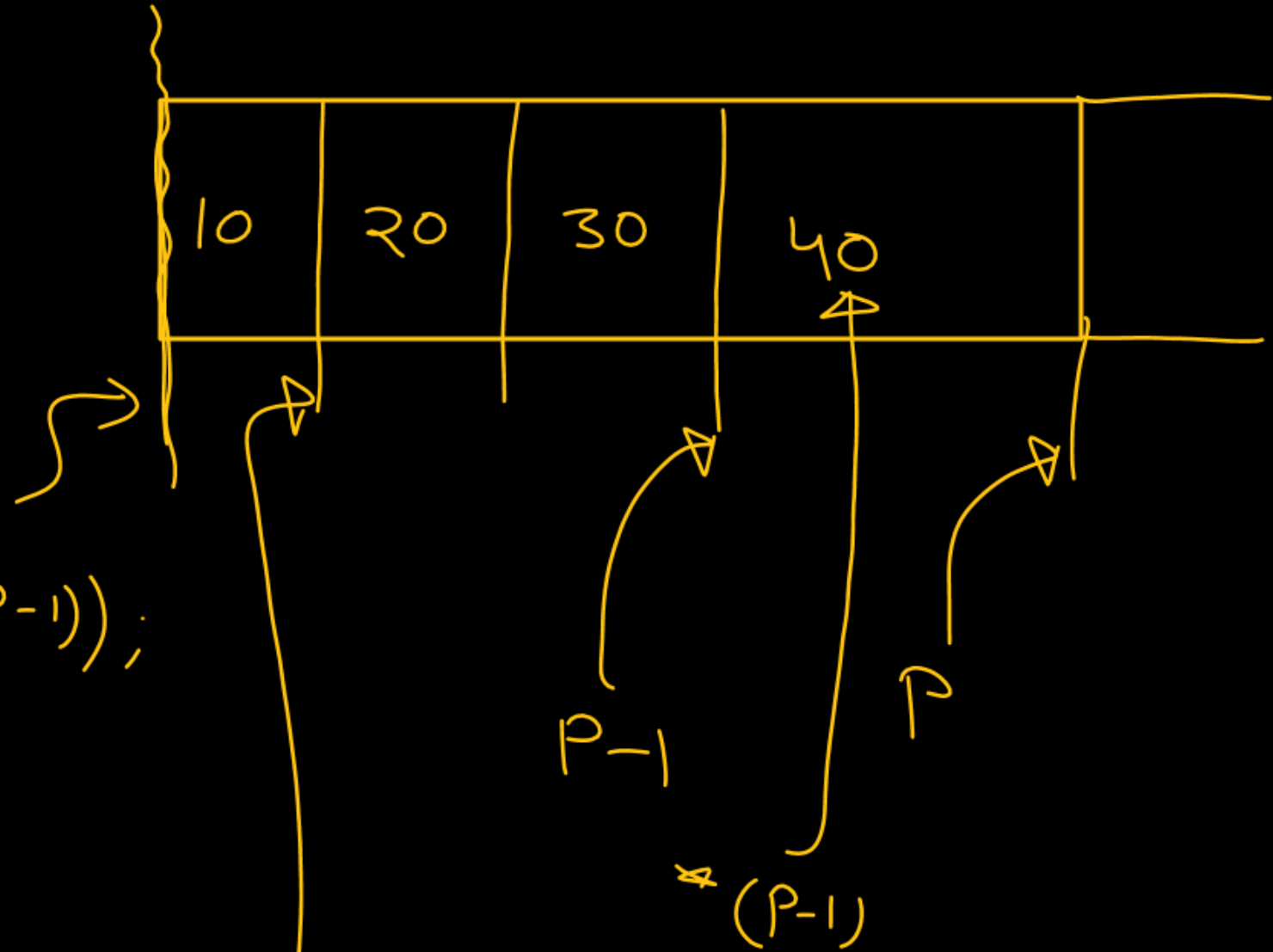
p = (int\*)(&a + 1); ✓

printf("%d %d", \*(a+1), \*(p-1));



a[1]

20 40



# Pointer to functions

```
void fun() {
```

```
    //
```

```
}
```

```
void add() {
```

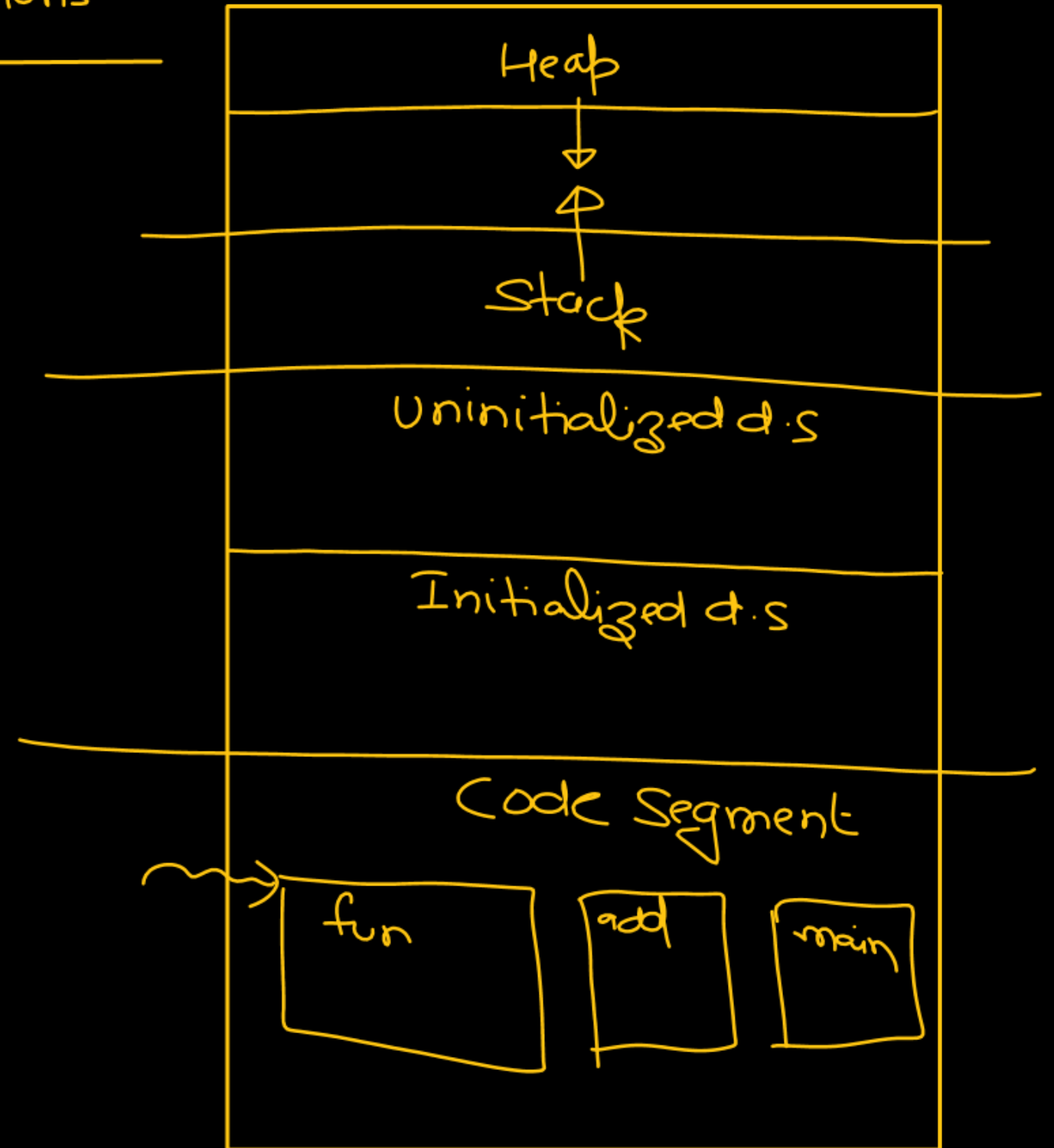
```
    //
```

```
}
```

```
void main() {
```

```
    //
```

```
}
```





int Add(int, int);

→ takes 2 <sup>int</sup> arg  
→ returns a integer

int (\*P)(int, int)

P is a pointer to function that takes 2 integer arguments  
and returns an integer.

```
#include <stdio.h>
```

```
int Add(int x, int y) {  
    int sum;  
    sum = x + y;  
    return sum;  
}
```

```
void main() {  
    int (*P)(int, int);  
    P = &Add;  
    printf("/d", (*P)(10, 20));  
}
```

P  $\Rightarrow$  &Add

Add(10, 20);

(\*P)(10, 20);

Add(10,20)

```
void main() {
```

```
    P = Add ; ✓
```

```
    (*P)(10,20); —
```

```
}
```

```
void main() {
```

```
    P = &Add ;
```

```
    (*P)(10,20);
```

```
}
```

```
P = Add
```

```
(P)(10,20);
```

```
P = Add
```

```
(*P)(10,20);
```



